

EMBEDDED SYSTEM AND DESIGN

EXPERIMENT NO. 3

CLOCK

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DISCUSSION

A clock is simply a device used to measure time. At present, it can be analog or digital, whichever people may prefer. One standard description of a clock is that it performs a regular movement of intervals determined by seconds, minutes, and hours. Clocks are integral to our daily lives, serving as consistent companions in people's journeys through time. With the provided details in mind, an Arduino kit, specifically the Arduino Uno, along with a 4-digit 7-segment display, four resistors, and five switch buttons, was employed to replicate a digital clock as proof of experiment 3.

The experiment could be divided into three parts: the continuous ticking of the clock, the section that increases the minutes or hours, and the part that decreases the minutes or hours. Since Arduino IDE does not allow access to real-time without an external module, the output

should have a timing of exactly 60000 milliseconds per tick, equivalent to one minute. At the same time, the clock's ticks should be increasing, which means that from 1:00, the following numbers should be 1:01, 1:02, 1:03, and so forth. Hence, with the use of an external library of *SevSeg.h* for the seven-segment display, the possible clock combinations were first stored in an array labeled *numarrgen()*, standing for number array generator, and the stored numbers would be displayed in the 7-segment with a delay of 60000 milliseconds. Additionally, the said delay was possible using the *millis()* function, which returns the number of milliseconds the program has been running. This function ensures the timing of a minute delay based on the program.

In setting up an accurate clock, one must be able to increase and decrease the minutes and hours with respect to the real-time. The concept behind increasing time adjustment starts with

pressing the fifth button. While the fifth button is held, the first four buttons are enabled. The first button, which handles the increment of hours, was coded in a way that will increase from 1 to 12 but will return to 1 when incremented once more. The second button, the handler for minute incrementing, adds the value of minutes from 0 to 59 and back to 0 again. The third button, which handles the decrement of hours, subtracts the value of hours from 12 to 1 and back to 12 again. Lastly, the fourth button is for the decrementing of minutes, which was patterned from 59 to 0 and back to 59. The logic in the code considers multiple if-else statements in verifying what button is pressed and the present value of the number being displayed. While the fifth button is pressed, adjusting the time is possible.

CONCLUSION

Overall, an actual clock was adequately recreated using the Arduino kit, showing the change in displayed values every minute and enabling time adjustment with switch buttons. As an application of embedded systems, the displaying of hours and minutes, together with their adjustment by incrementing and decrementing, using the seven-segment display and button switches, were employed. By configuring the state of the 4-digit seven-segment display, applying the circuit's function of time with delays and switches, and showing the appropriate clock features through an embedded system, the lessons in microprocessors, logic circuits, and switching theory were applied. Thus, as a requirement of the embedded system and design, a clock was effectively constructed, thereby satisfying all objectives.